Homework 3 – Motion Estimation & Compensation

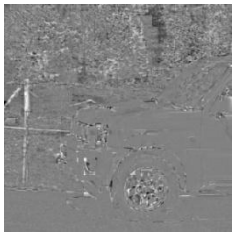
314552046 莊明憲

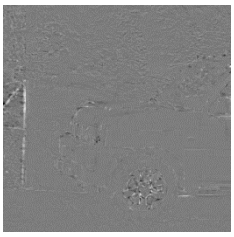
SPEC

- Motion estimation (ME)
  - Block size: 8x8
  - Search range: [+-8]
  - Full search block matching algorithm
  - Integer precision
- Motion compensation (MC)
  - Save the reconstructed frame (after MC) and the residual
- Search range:
  - Compare the results (in PSNR and runtime) with different search ranges ([+-8], [+-16], [+-32]).
- Three-step search
  - Compare the results (in PSNR and runtime) with the Full search algorithm.

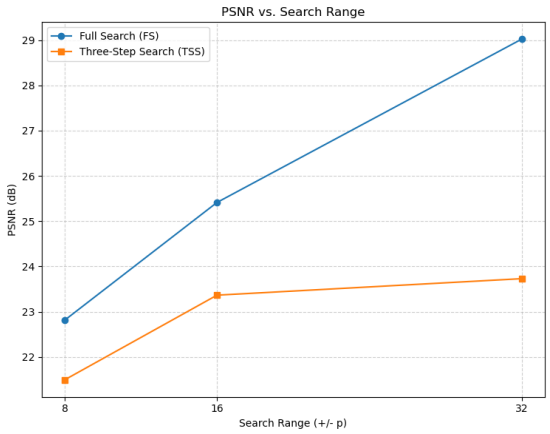
Results

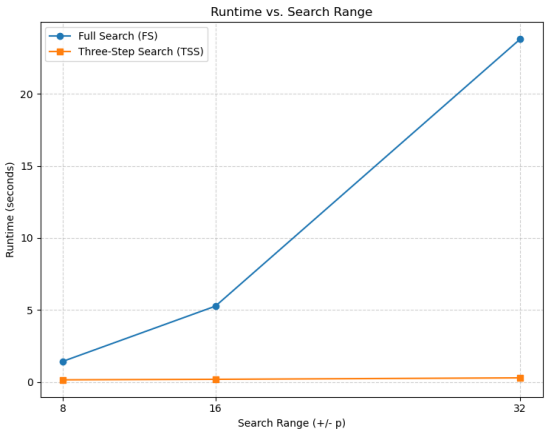
Search Range : 8			
Full Search Reconstructed	Full Search Residual	Three Step Search Reconstructed	Three Step Search Residual

Search Range : 16			
Full Search Reconstructed	Full Search Residual	Three Step Search Reconstructed	Three Step Search Residual
			

Search Range : 32			
Full Search Reconstructed	Full Search Residual	Three Step Search Reconstructed	Three Step Search Residual
			

Comparison





	Search Range : 8		Search Range : 16		Search Range : 32	
	PSNR(dB)	Runtime(s)	PSNR(dB)	Runtime(s)	PSNR(dB)	Runtime(s)
Full Search	22.811	1.446	25.412	5.270	29.019	23.793
Three Step Search	21.494	0.156	23.364	0.192	23.729	0.294
TSS vs. FS	−1.317 dB	9.27x faster	−2.048 dB	27.44x faster	− 5.291 dB	80.93x faster

1. Full Search (FS) Performance

The PSNR for Full Search significantly improves as the search range expands, rising from 22.811 dB (p=8) to 29.019 dB (p=32). This strong correlation suggests the video sequence contains high motion, as better matches are consistently found at greater distances. As expected, the runtime grows dramatically from 1.446 s to 23.793 s. This behavior is due to the algorithm's

O(p^2) complexity, which requires checking every possible candidate block in the expanding search area.

2. Three-Step Search (TSS) Performance

In contrast, the PSNR for TSS stagnates quickly after p=16, peaking at only 23.729 dB. This demonstrates the algorithm's primary limitation, as it gets "lost" and settles for a local minimum when the true motion vector lies far from the center. The main advantage of TSS is its speed, with runtime remaining nearly constant (0.156 s to 0.294 s). This is because TSS performs a fixed, low number of comparisons regardless of the search range.

3. Conclusion and Trade-off

This high-motion video clearly illustrates the algorithm trade-off. TSS is significantly faster (up to 80.93x at p=32), but this speed comes at a severe quality cost, losing 5.291 dB of PSNR. For this sequence, TSS is an unsuitable choice as it fails to track the large motion. While computationally expensive, Full Search is the only method that successfully finds the correct motion vectors, proving its necessity for high-motion scenarios.